Anderson localisation with self-avoiding-walks

Giulia Liconti

Geboren am 21. November 2004 in Bologna

22nd August 2026

Bachelor's Thesis Mathematics

Advisor: Prof. Dr. Margerita Disertori

Second advisor: Dr. Davide Macera

Institut of applied Mathematics

Mathematisch-Naturwissenschaftliche Fakultät der

Rheinischen Friedrich-Wilhelms-Universität Bonn

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analougue of a random walk in a random environment. The model considers a d-dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump , it indicates the absence of quantum transport.This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.
Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.
The two important methods to prove Anderson localization are the *multiscale analysis method* (MSA), developed by Froehlich and Spencer [] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. We will deal with the second one, which is more elementary and gives stronger results on *dynamical localization* [Chapter 2.2].

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic concepts such as indipendence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schroedinger operator. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix and V_ω is a diagonal matrix with random entries. This operator acts on the Hilbert space

$$l^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \to \mathbb{C} : \sum_{n \in \mathbb{Z}^d} |u(n)|^2 < \infty\}$$

with inner product $\langle u, v \rangle = \sum_n \overline{u(n)} v(n)$.

1.1.1 The operator

The Descrete Laplacian. The kinetic term $-\Delta$ is the descrete analogue of the usual negative Lalacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in l^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := - \sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k) \quad \text{for } x \in \mathbb{Z}^d.$$

where $| * |$ denotes the l^2 norm.

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For each site x, there are $2d$ possible k such that $|x - k| = 1$. We can now define an infinte matrix such that:

$$(-\Delta)_{x,k} = \begin{cases} 2d & \text{if } x = k \quad \text{(on the diagonal)} \\ -1 & \text{if } |x-k| = 1 \quad (k \sim x, \text{ first neighbors}) \\ 0 & \text{otherwise} \end{cases}$$

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2, bounded, and self-adjoint.

The first claim (summability) is shown by:

$$\begin{aligned} \sum_{x\in\mathbb{Z}^d} |\Delta u(x)|^2 &= \sum_{x\in\mathbb{Z}^d} \left| \sum_{k\sim x, k\in\mathbb{Z}^d} \big(u(x) - u(k)\big) \right|^2 \\ &\leq 2d \sum_{x\in\mathbb{Z}^d} \left(\sum_{k\sim x} |u(x) - u(k)|^2 \right) \\ &\leq 2d \sum_{x\in\mathbb{Z}^d} \left(2 \sum_{k\sim x, k\in\mathbb{Z}^d} |u(x)|^2 + |u(k)|^2 \right) \\ &\leq 2d \cdot 2d \cdot 4\|u\|_2^2 < \infty. \end{aligned}$$

In each step we used, in order, Cauchy-Schwarz, the bound over the number of neighbors, triangle inequality and the invariance under translation of the l^2 norm.

The boundedness comes from:

$$\frac{\|\Delta u\|_2^2}{\|u\|_2^2} \leq 16d^2 \implies \|\Delta\|_2 \leq 4d.$$

To prove self-adjointness, we only need to show symmetry (given $-\Delta$ is bounded). For $u, v \in l^2(\mathbb{Z}^d)$ and $k \in \mathbb{Z}^d$:

$$\langle u, \Delta v\rangle = \sum_{x\in\mathbb{Z}^d} \overline{u(x)} \Delta v(x) = \sum_{x\in\mathbb{Z}^d} u(x) \left(\sum_{k\sim x} v(x) - v(k) \right)$$

$$= \sum_{x\in\mathbb{Z}^d} \left(u(x) 2d v(x) - \sum_{k\sim x} u(x) v(k) \right) = \sum_{x\in\mathbb{Z}^d} 2d u(x) v(x) - \sum_{x\in\mathbb{Z}^d} \sum_{k\sim x} u(k) v(x)$$

$$= \sum_{x\in\mathbb{Z}^d} v(x) \left(\sum_{k\sim x} u(x) - u(k) \right) = \sum_{x\in\mathbb{Z}^d} \overline{v(x)} \Delta u(x) = \langle \Delta u, v\rangle.$$

The spectrum of the bounded, self-adjoint operator $-\Delta$ is the closed, real-valued interval $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns.

In order to understand this results better, let's consider the discrete Fourier transform:

$$\mathcal{F}\colon \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0,2\pi)^d)$$

such that:

$$(\mathcal{F}u)(x) \coloneqq \lim_{N\to\infty} (2\pi)^{-\frac{d}{2}} \sum_{|n|\le N} u(n)\, e^{-ix\cdot n}, \qquad x \in [0,2\pi)^d.$$

For semplicity of the calculations we don't consider the limit and just sum over all $n \in \mathbb{Z}^d$ and u summable.

$$\begin{aligned}
(\mathcal{F}\Delta u)(x) &= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} (\Delta u)(n)\, e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(\sum_{|k-n|=1} \big(u(n) - u(k)\big) \right) e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(2d\, u(n) - \sum_{|k-n|=1} u(k) \right) e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \sum_{|k-n|=1} u(k)\, e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k) \sum_{|m|=1} e^{-ix\cdot(k+m)} \quad (\text{for } m = n-k) \\
&= 2d\,(\mathcal{F}u)(x) - \left((2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k)\, e^{-ix\cdot k} \right) \sum_{|m|=1} e^{-ix\cdot m} \\
&= 2d\,(\mathcal{F}u)(x) - (\mathcal{F}u)(x) \sum_{j=1}^{d} \big(e^{-ix_j} + e^{ix_j}\big) \\
&= \left(2d - 2\sum_{j=1}^{d} \cos(x_j) \right) (\mathcal{F}u)(x) \\
&= \left(2\sum_{j=1}^{d} 1 - \cos(x_j) \right) (\mathcal{F}u)(x).
\end{aligned}$$

We obtain the multiplication operator:

$$\mathcal{F}\Delta\mathcal{F}^{-1} = 2\sum_{j=1}^{d} 1 - \cos(x_j)$$

on $L^2([0,2\pi)^d)$ in the variable $x = (x_1, \ldots.., x_d)$. We notice that the Laplacian doesn't have eigenvalues in $l^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $x \in [0,2\pi)^d$ and set

$$\phi_x(n) := e^{-in\cdot x}$$

While $\phi_x \notin l^2(\mathbb{Z}^d)$, Δ acts on it as

$$\begin{aligned}(\Delta\phi_x)(n) &= \sum_{|k|=1} e^{-in\cdot x} - e^{-i(n+k)\cdot x} \\ &= 2d\phi_x(n) - \left(\sum_{j=1}^{d} 2\cos(x_j)\right)\phi_x(n) \quad = \left(2\sum_{j=1}^{d} 1 - \cos(x_j)\right)\phi_x.\end{aligned}$$

Thus ϕ_x is a bounded generalized eigenfunction of h_0 to the spectral value $-2\sum_j \cos(x_j)$.

The descrete Laplacian, doesn't have eigenfunction in $l^2(\mathbb{Z}^d)$, but has continuous plane waves as generalized eigenfunctions. ...

The Potential Term.

Definition 1.1 (i.i.d. random variables)**.** $(\omega_v)_{v\in V}$ are identically distributed if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_v \in A) = \mu(A)$ for all $v \in V$ and Borel sets $A \subset \mathbb{R}$. $(\omega_v)_{v\in V}$ are independent if

$$\mathbb{P}(\omega_{v_1} \in A_1, \ldots, \omega_{v_l} \in A_l) = \prod_{k=1}^{l} \mathbb{P}(\omega_{v_k} \in A_k) = \prod_{k=1}^{l} \mu(A_k)$$

for each finite subset $\{v_1, \ldots, v_l\}$ of V and arbitrary Borel sets $A_1, \ldots, A_l \subset \mathbb{R}$.

We assume that the distribution μ of ω_v is absolutely continuous with density ρ where ρ is bounded with compact support, i.e.,

$$\mu(B) = \int_B \rho(u)\,du \quad \text{for } B \subset \mathbb{R} \text{ Borel}, \quad \rho \in L_0^\infty(\mathbb{R}).$$

Physically, V_ω represents the random electric potential created by nuclei at the sites $v \in \mathbb{Z}^d$. ... **ovviamente il testo da modificare in base al resto delle notazioni .**
$\lambda > 0$ is the disorder parameter

1.1.2 The spectrum of H_ω.

- in $\mathbb{R}$
- bounded $[0, 4d] + \text{supp}\mu$
- spectral division (any use of the spectral theorem should be sent to the Appendix)
- RAGE theorem (or maybe directly in the Appendix or chapter 2)

Dalla teoria spettrale per operatori autoaggiunti su uno spazio di Hilbert $\mathcal{H}$, lo spazio si decompone univocamente nella somma diretta di sottospazi invarianti ortogonali:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_{sc} \oplus \mathcal{H}_{ac} \tag{1.1}$$

a cui corrispondono rispettivamente lo spettro puramente puntuale $\sigma_{pp}(H)$, singolarmente continuo $\sigma_{sc}(H)$ e assolutamente continuo $\sigma_{ac}(H)$, con $\sigma(H) = \sigma_{pp}(H) \cup \sigma_{sc}(H) \cup \sigma_{ac}(H)$.

L'interpretazione fisica di questi sottospazi è chiarita dal fondamentale **Teorema RAGE** (Ruelle-Amrein-Georgescu-Enss):

Theorem 1.2 (RAGE)**.** *Sia H un operatore di Schrödinger su $l^2(G)$ e $\mathcal{H}_c = \mathcal{H}_{ac} \oplus \mathcal{H}_{sc}$ il sottospazio spettrale continuo. Allora:*

1. *$\phi \in \mathcal{H}_{pp}$ se e solo se per ogni $\epsilon > 0$ esiste un raggio $R < \infty$ tale che:*

$$\sup_{t\in\mathbb{R}} \|\chi_{(|x|>R)} e^{-itH}\phi\| < \epsilon \quad \Longleftrightarrow \quad \lim_{R\to\infty} \sup_{t\in\mathbb{R}} \|\chi_{(|x|>R)} e^{-itH}\phi\| = 0$$

ovvero lo stato ϕ rimane confinato in regioni compatte uniformemente nel tempo (stato legato).

2. *$\psi \in \mathcal{H}_c$ se e solo se per ogni $R > 0$:*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \|\chi_{(|x|\leq R)} e^{-itH}\psi\| \, dt = 0$$

ovvero la probabilità che lo stato permanga in qualsiasi regione limitata decade in media temporale a zero (stati di scattering / trasporto).

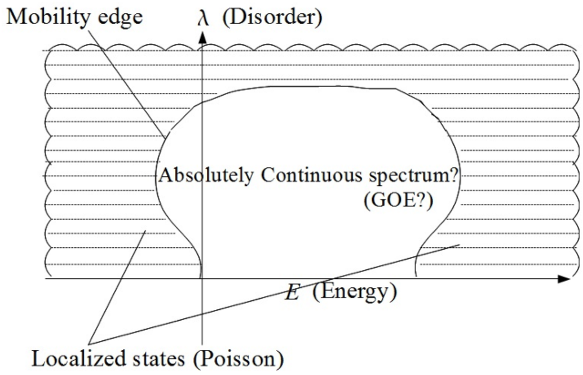

Figure 1: Aizenman figure

1.2 Main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:
The aim of this thesis is to prove localisation on the whole spectrum of H_ω ...
Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.
potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei rislutati con Frohlich e spencer e con Aizenman e Molchanov.
One of the main results concerns the values of λ

- $\lambda >> 1$ and $\lambda = 0$
- general case for $d = 1, d \geq 2$

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 spectral localisation

Definition 2.1 (Spectral Localization)**.** Sia $I \subset \mathbb{R}$ un intervallo aperto. Un operatore di Schrödinger casuale H_ω presenta **localizzazione spettrale** in I se, con probabilità 1 (quasi certamente), lo spettro continuo in I è vuoto e lo spettro è puramente puntuale:

$$\sigma_c(H_\omega) \cap I = \emptyset \quad \text{e} \quad \sigma(H_\omega) \cap I \subset \sigma_{pp}(H_\omega) \quad \text{quasi certamente.}$$

Definition 2.2 (Exponential Spectral Localization)**.**

$$|\phi_{\omega,n}(x)| \leq C_{\omega,n} e^{-\mu|x - x_{n,\omega}|} \tag{2.1}$$

Remark 2.3 (Spettro puntuale denso). closure set of eigenvalues

2.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t) = \left\| \, |x| \, e^{-itH_\omega} P_{H_\omega \in [a,b]} \phi_0 \right\|^2 \tag{5}$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av}, then roughly $x(t) \sim v_{av} t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ will be proportional to t^2 for large t.

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ is bounded uniformly in t. It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t \to \infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0, as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t \to \pm\infty} \frac{\langle x^2 \rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \qquad \text{for all } \delta > 0, \tag{6}$$

for a large class of compactly supported initial conditions ϕ_0. *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n, le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

2.2 dynamical localisation

Definition 2.4.

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k\rangle|\right) \leq Ce^{-\mu|j-k|}$$

for all φ with compact support.

further explaination of the meaning

2.2.1 finite moments

Proposition: *Dynamical localization implies that all moments of the position operator are bounded in time, i.e., for all $p > 0$ and all finitely supported $\psi \in \ell^2(\mathbb{Z}^d)$,*

$$\sup_{t\in\mathbb{R}} \||X|^p e^{-ith_\omega}\psi\| < \infty \quad \textit{almost surely,} \tag{2.2}$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$.

Proof. To see how (1) follows from dynamical localization, assume that $\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned}
\||X|^p e^{-ith_\omega}\psi\|^2 &= \sum_j |\langle e_j, |X|^p e^{-ith_\omega}\psi\rangle|^2 \\
&= \sum_j |j|^{2p} \left| \sum_{|k|\leq R} \langle e_j, e^{-ith_\omega} e_k\rangle \psi(k)\right|^2 \\
&\leq \sum_j \sum_{|k|\leq R} |j|^{2p} |\langle e_j, e^{-ith_\omega} e_k\rangle|^2 \|\psi\|^2,
\end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-ith_\omega} e_k\rangle|^2$ (as this number is bounded by 1) and then take expectations to get

$$
\begin{aligned}
\mathbb{E}\left(\sup_t \||X|^p e^{-ith_\omega}\psi\|^2\right) &\leq \sum_j \sum_{|k|\leq R} |j|^{2p}\mathbb{E}\left(\sup_t |\langle e_j, e^{-ith_\omega} e_k\rangle|\right) \|\psi\|^2 \\
&\leq C \sum_j \sum_{|k|\leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\
&< \infty.
\end{aligned}
$$

□

2.3 From dynamical to spectral localisation

Theorem 2.5 (Localizzazione Dinamica $\implies$ Localizzazione Spettrale)**.** *Se H_ω esibisce localizzazione dinamica in un intervallo aperto I, allora H_ω presenta quasi certamente solo spettro puramente puntuale in I ($\sigma_c(H_\omega)\cap I = \emptyset$).*

Schema della dimostrazione via RAGE. Applicando il Teorema RAGE alla proiezione continua $P_{cont}(H)\psi$ per uno stato ψ a supporto compatto in $\{|y| \leq r\}$:

$$
\mathbb{E}(\|P_{cont}(H)\psi\|^2) \leq \lim_{R\to\infty} \frac{1}{T}\int_0^T dt \sum_{|x|\geq R, |y|\leq r} \mathbb{E}(|\langle \delta_x, e^{-itH}\delta_y\rangle|)\|\psi\|^2
$$

Per localizzazione dinamica, la somma $\sum_{|x|\geq R} e^{-\mu|x-y|}$ tende a zero quando $R \to \infty$. Dunque $P_{cont}(H)\psi = 0$ quasi certamente su un insieme denso di stati ψ, da cui $\sigma_c(H_\omega) \cap I = \emptyset$. □

3 Fractional moments

3.1 The green function

The *resolvent operator* is defined as:

$$G(z) = (H - z)^{-1}$$

3.2 Localisation at large disorder

Theorem 3.1 (Aizenman–Molchanov, 1993). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}\left(|G_{\omega,\lambda}(x, y; z)|^s\right) \leq Ce^{-\mu|x-y|} \tag{3.1}$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$.

Lemma 3.2 (A priori bound). *There exists a constant $C_1 = C_1(s, \rho) < \infty$ such that*

$$\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x, y; z)|^s) \leq C_1\lambda^{-s} \tag{3.2}$$

for all $x, y \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Here

$$\mathbb{E}_{x,y}(\dots) = \int\int \dots \rho(\omega_x)\, d\omega_x \rho(\omega_y)\, d\omega_y$$

is the conditional expectation with $(\omega_u)_{u\in\mathbb{Z}^d\setminus\{x,y\}}$ fixed. After averaging over ω_x and ω_y the bound in (1) does not depend on the remaining random parameters. Thus we get also that

$$\mathbb{E}(|G_{\omega,\lambda}(x, y; z)|^s) \leq C_1\lambda^{-s}.$$

Proof. We first prove (1) for the case $x = y$, which demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u\in\mathbb{Z}^d\setminus\{x\}}$. With $P_{e_x} := \langle e_x, \cdot\rangle e_x$, the orthogonal projection onto the span of e_x, we can separate the ω_x and $\hat{\omega}$ dependence of $H_{\omega,\lambda}$ as

$$H_{\omega,\lambda} = H_{\hat{\omega},\lambda} + \lambda\omega_x P_{e_x}.$$

The resolvent identity yields

$$(H_{\omega,\lambda} - z)^{-1} = (H_{\hat{\omega},\lambda} - z)^{-1} - \lambda\omega_x H_{\hat{\omega},\lambda} - z)^{-1} P_{e_x}(H_{\omega,\lambda} - z)^{-1}. \tag{3.3}$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_{\omega,\lambda}(x,x;z) = G_{\hat{\omega},\lambda}(x,x;z) - \lambda\omega_x G_{\hat{\omega},\lambda}(x,x;z) G_{\omega,\lambda}(x,x;z) \tag{3.4}$$

or

$$G_{\omega,\lambda}(x,x;z) = \frac{1}{a+\lambda\omega_x} \quad \text{with } a = \frac{1}{G_{\hat{\omega},\lambda}(x,x;z)}. \tag{3.5}$$

Note that the latter is well-defined since one can easily check the Herglotz property $\operatorname{Im} G_{\hat{\omega},\lambda}(x,x;z)/\operatorname{Im} z > 0$ of the Green function.

The important fact is that a is a complex number which does not depend on ω_x. Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x)\, d\omega_x$, we find that

$$\mathbb{E}_x(|G_{\omega,\lambda}(x,x;z)|^s) \leq \frac{\|\rho\|_\infty}{\lambda^s} \int_{\operatorname{supp}\rho} \frac{d\omega_x}{|\frac{a}{\lambda}+\omega_x|^s} \leq \frac{C(\rho,s)}{\lambda^s}, \tag{3.6}$$

with $C(\rho,s)$ independent of λ and a, and thus independent of $\hat{\omega}$, z and x.

potrei fare una piccola parentesi in cui dico che per distribuzione uniforme prende un valore specifico.

The proof of (1) for $x \neq y$ is based on the same idea, replacing the rank-one-perturbation arguments above with rank-two-perturbation arguments. Write $\omega = (\hat{\omega}, \omega_x, \omega_y)$, $P = P_{e_x} + P_{e_y}$ and

$$H_{\omega,\lambda} = H_{\hat{\omega},\lambda} + \lambda\omega_x P_{e_x} + \lambda\omega_y P_{e_y}.$$

Using the resolvent identity similar to above one arrives at

$$P(H_{\omega,\lambda}-z)^{-1}P = \left(A + \lambda\begin{pmatrix}\omega_x & 0\\ 0 & \omega_y\end{pmatrix}\right)^{-1}, \tag{3.7}$$

where

$$A = \left(P(H_{\hat{\omega},\lambda}-z)^{-1}P\right)^{-1},$$

both to be read as identities for 2×2-matrices in the range of P. This is a special case of the Krein formula which characterizes the resolvents of finite-rank perturbations of general self-adjoint operators. For the matrix A one can check that $\operatorname{Im} A = \frac{1}{2i}(A-A^*) < 0$ if $\operatorname{Im} z > 0$ and $\operatorname{Im} A > 0$ if $\operatorname{Im} z < 0$. It is also independent of ω_x and ω_y.

Using that $G_{\omega,\lambda}(x,y;z)$ is one of the matrix-elements of $P(H_{\omega,\lambda}-z)^{-1}P$, we find

$$\begin{aligned}
\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x,y;z)|^s) &\le \mathbb{E}_{x,y}\left(\left\|\left(A+\lambda\begin{pmatrix}\omega_x & 0\\ 0 & \omega_y\end{pmatrix}\right)^{-1}\right\|^s\right)\\
&= \lambda^{-s}\mathbb{E}_{x,y}\left(\left\|\left(-\frac{1}{\lambda}A-\begin{pmatrix}\omega_x & 0\\ 0 & \omega_y\end{pmatrix}\right)^{-1}\right\|^s\right)\\
&\le \frac{\|\rho\|_\infty^2}{\lambda^s}\int_{-r}^{r}\int_{-r}^{r}\left\|\left(-\frac{1}{\lambda}A-\begin{pmatrix}\omega_x & 0\\ 0 & \omega_y\end{pmatrix}\right)^{-1}\right\|^s d\omega_x\, d\omega_y,
\end{aligned}$$

where $[-r,r]$ is an interval containing $\operatorname{supp}\rho$. In the double integral we change variables to

$$u=\frac{1}{2}(\omega_x+\omega_y),\quad v=\frac{1}{2}(\omega_x-\omega_y),$$

which gives a Jacobian factor of 2. As $(\omega_x,\omega_y)\in[-r,r]^2$ implies $(u,v)\in[-r,r]^2$ we arrive at the bound

$$\begin{aligned}
\mathbb{E}_{x,y}(|G_{\omega,\lambda}(x,y;z)|^s) &\le \frac{2\|\rho\|_\infty^2}{\lambda^s}\int_{-r}^{r}\int_{-r}^{r}\left\|\left(-\frac{1}{\lambda}A+\begin{pmatrix}-v & 0\\ 0 & v\end{pmatrix}-uI\right)^{-1}\right\|^s du\, dv\\
&\le \frac{4r\|\rho\|_\infty^2}{\lambda^s}C(r,s)=\frac{C(s,\rho)}{\lambda^s}.
\end{aligned}$$

That the latter bound is uniform in x,y and z as well as in the random parameters $(\omega_u)_{u\in\mathbb{Z}^d\setminus\{x,y\}}$ follows from the fact that the matrix

$$-\frac{1}{\lambda}A+\begin{pmatrix}-v & 0\\ 0 & v\end{pmatrix}$$

has either positive or negative imaginary part and the following general result:

For every $s\in(0,1)$ and $r>0$ there exists $C(r,s)<\infty$ such that

$$\int_{-r}^{r}\|(B-uI)^{-1}\|^s\,du\le C(r,s) \tag{3.8}$$

for all 2×2-matrices B such that either $\operatorname{Im}B\ge 0$ or $\operatorname{Im}B\le 0$.

Starting with the observation that, by Schur's Theorem, B may be assumed upper triangular, we also may assume without loss that $\operatorname{Im}B\ge 0$.

Thus

$$B = \begin{pmatrix} b_{11} & b_{12} \\ 0 & b_{22} \end{pmatrix} \tag{3.9}$$

and

$$(B - uI)^{-1} = \begin{pmatrix} \frac{1}{b_{11}-u} & -\frac{b_{12}}{(b_{11}-u)(b_{22}-u)} \\ 0 & \frac{1}{b_{22}-u} \end{pmatrix}. \tag{3.10}$$

The bound (7) follows if we can establish a corresponding fractional integral bound for the absolute value of each entry of (9) separately. For the diagonal entries this is obvious.

We bound the upper right entry of (9) by

$$\left| \frac{b_{12}}{(b_{11}-u)(b_{22}-u)} \right| \le \frac{|b_{12}|}{|\operatorname{Im}((b_{11}-u)(b_{22}-u))|} = \frac{1}{\left| u \frac{\operatorname{Im} b_{11} + \operatorname{Im} b_{22}}{|b_{12}|} - \frac{\operatorname{Im}(b_{11} b_{22})}{|b_{12}|} \right|}. \tag{3.11}$$

The positive matrix

$$\operatorname{Im} B = \begin{pmatrix} \operatorname{Im} b_{11} & \frac{1}{2i} b_{12} \\ -\frac{1}{2i} \bar{b}_{12} & \operatorname{Im} b_{22} \end{pmatrix}$$

has positive determinant, i.e. $\det \operatorname{Im} B = \operatorname{Im} b_{11} \operatorname{Im} b_{22} - |b_{12}|^2/4 \ge 0$. We thus get

$$\left| \frac{\operatorname{Im} b_{11} + \operatorname{Im} b_{22}}{b_{12}} \right|^2 \ge \frac{2 \operatorname{Im} b_{11} \operatorname{Im} b_{22}}{|b_{12}|^2} \ge \frac{1}{2}.$$

The latter allows to conclude the required integral bound for (10). $\square$

3.2.1 Proof without SAW

4 SAW representation

4.1 SAW Definitions

1.

4.2 SAW representation of G

- Theorem of the representation

    1. first prove that $G(x,y) = G(x,x) \sum G(x_1, y)$
    2. then the represantion with the R_N

- put figure of the grid ant the different paths

4.3 Proof with SAW

5 From fractional moments to dynamical localisation

Theorem 5.1. *Theorem final*

Proposition 5.2.

$$|\Im z|\mathbb{E}\left(|G(x,y;z)|^2\right) \leq C_1\mathbb{E}\left(|G(x,y;z)|^s\right)$$

Proof of theorem 5.1. □